REPUBLIC OF THE PHILIPPINES
DEPARTMENT OF INFORMATION AND COMMUNICATIONS TECHNOLOGY

PHILIPPINE STARTUP CHALLENGE XI

Department of Information and Communications Technology

CONCEPT NOTE

WasteLedger PH

Every kilo counted, paid for, and proven.

Your food waste is worth something. WasteLedger PH helps Philippine cities pay people for sorting it, so it becomes compost instead of landfill. Starting in Baguio City.

Team T+N

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I. SUMMARY

What it is. WasteLedger PH is software that lets a Philippine city pay public market vendors and farmers a small amount for each kilo of clean vegetable and food waste they hand over, so the waste is composted instead of trucked to a landfill. Think of a junk shop for food waste: the city pays, out of what it saves on hauling, and every kilo is weighed and checked before anyone is paid.

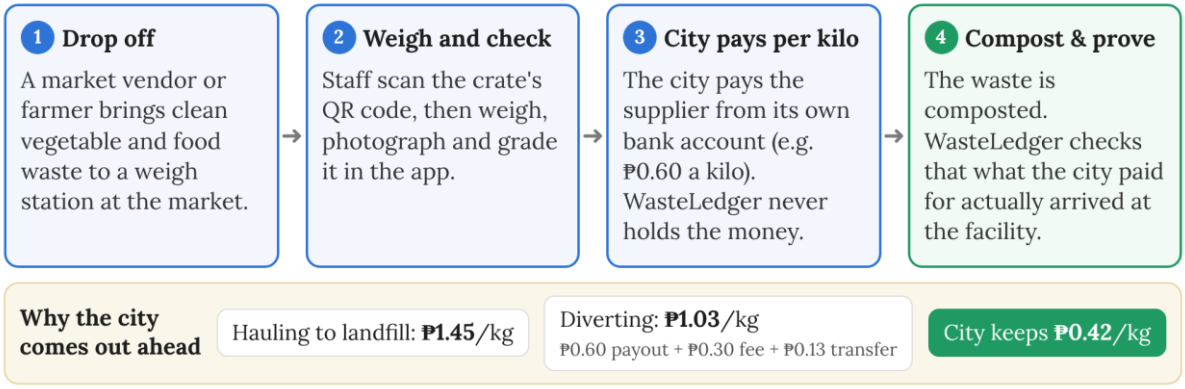


Figure 1. How WasteLedger works, using Baguio's costs (before staff costs). Source: GSO (2026); team computation.

The problem. Philippine cities are running out of places to put their garbage and are paying more to move it. Only 805 of the 1,592 local government units (LGUs) tracked by the Department of Environment and Natural Resources (DENR) are served by a sanitary landfill ("DENR: PH nears," 2025; Virador & Paluga, 2026). After part of Cebu City's Binaliw landfill collapsed in January 2026, killing 36 people, the city's daily hauling bill doubled to ₱4.2 million (Virador & Paluga, 2026). About half of all municipal waste is biodegradable (Mantaring, 2024), and public markets are among its largest sources, yet the team found no Philippine program that pays per kilo for this waste and checks that it reaches a composting facility.

Why start in Baguio. Baguio has no landfill of its own. In 2025 it trucked 80,472 tons of residual waste to a landfill in Urdaneta, Pangasinan, for about ₱127.7 million (General Services Office [GSO], 2026), and the DENR now lists that landfill as in imminent danger (Virador & Paluga, 2026). The city already composts some market waste at its Irisan facility, which has spare capacity, and its GSO has shared the real cost data used here. Because Baguio already composts part of its market waste, savings measured here are a floor for cities that compost none.

Why the city saves money. Hauling and dumping one kilo of waste costs Baguio about ₱1.45. Diverting it costs about ₱1.03: a ₱0.60 payout to the supplier, a ₱0.30 WasteLedger fee and up to ₱0.13 in transfer fees. The city keeps about ₱0.42 per kilo before staff costs; a new station worker's wage is covered once a station handles about 1.2 tons a day, and below that the city can use market or GSO utility workers who already handle this waste. The same software works in any city; only local settings, such as the hauling cost and the composting facility, change.

Stage and ask. Ideation stage. The model and the Baguio baseline are defined, and during PSC XI the team will build a clickable prototype. The GSO has shared its data, but no city has yet agreed to run or fund a pilot; the team is requesting letters of interest. A three-month pilot costs about ₱119,000 to ₱217,000. The team seeks about ₱45,000 to ₱96,000 for equipment, software, registration and contingency, about ₱26,000

in bridge funding for supplier payouts until a city can fund them, and mentorship on payments, data privacy and local government procurement.

In one line: every kilo counted, paid for, and proven.

II. BACKGROUND OF THE PROBLEM

A. A national problem

The Philippines produces about 14.6 million tons of municipal waste a year, about half of it biodegradable (Dela Peña, 2025; Mantaring, 2024). Most LGUs now have a solid waste plan, 1,416 of 1,592 by 2025 ("DENR: PH nears," 2025); the bottleneck is disposal. Only 805 LGUs are served by one of the country's 397 sanitary landfills, and after the Binaliw collapse the DENR found five landfills in imminent danger (Virador & Paluga, 2026). The largest cities feel it most:

City	Waste per day	What is happening
Quezon City	about 3,600 t	Spent ₱2.1 billion on environmental and sanitary services in 2024 ("Tracking trash costs," 2026); its zero-waste public market composts only small volumes so far (Quezon City Government, n.d.)
Manila	about 1,174 t	Raised business garbage fees, in one reported case twelvefold, after its waste was redirected to a landfill in Rizal (Lalu, 2026)
Caloocan	about 912 t	Its spending on environmental and sanitary services rose in 2024 ("Tracking trash costs," 2026)
Davao City	about 753 t	Still using a landfill that is at full capacity ("Davao City to continue," 2025)
Cebu City	about 700 t	Binaliw landfill collapsed in January 2026; waste now goes 61 km away at ₱3,906 a ton, about double the city's usual cost (Virador, 2026)

Waste per day as reported by the DENR ("Waste-to-energy," 2025; "Waste volume," 2025) and, for Davao City, the 2025 average ("Davao City to continue," 2025). For scale only.

B. Baguio: what it hauls and what it costs

Baguio generates 550 to 593 tons of waste a day, and its 2022 waste analysis and characterization study (WACS) found 35.9% of it biodegradable (Camila et al., 2026; Cimatu, 2025). Residual waste goes from the Marcos Highway transfer station to a sanitary landfill in Urdaneta, Pangasinan (GSO, 2026).



Figure 2. Residual waste hauled out each month (left) and the hauling plus tipping cost per ton (right), 2025. Source: GSO (2026); team computation.

The lower 2025 rate is not secure: the Urdaneta landfill was reported almost full in May 2025 (Micua, 2025) and is now on the DENR's imminent-danger list (Virador & Paluga, 2026).

C. The gap: no one pays for, or checks, the handoff



Figure 3. Baguio's public markets produced 15,511 tons of waste in 2025; only the vegetable waste (2,551 tons, about 16%) went to the Irisan Composting Facility, which has spare capacity. The GSO puts Irisan's typical intake at about 5 tons a day; the 2025 total averages about 7. Source: GSO (2026).

Irisan already turns market waste into compost, so the weak link is not composting but the handoff: no one weighs each vendor's food waste, pays vendors to sort it cleanly, or checks it against what the facility

receives. Upstream, Benguet farmers lose 20% to 50% of their produce in marketing (Sanchez et al., 2019). Baguio has also committed, under its Open Government Partnership plan, to a household waste-tracking app with incentives, but has set no milestones (Open Government Partnership [OGP], n.d.).

The GSO says market food waste already goes to Irisan, so little may be left in the garbage sent to the landfill. No one has measured it yet; Section VII shows the savings at 10%, 20% and 35.9%, and the pilot will measure it.

D. Policy direction

Policy	What it asks	How WasteLedger helps
RA 9003	Segregation and at least 25% diversion; LGU self-compliance reports	Diversion records per supplier, ready for reports (Ecological Solid Waste Management Act of 2000, 2001; National Solid Waste Management Commission [NSWMC], 2016)
Seal of Good Local Governance, Department of the Interior and Local Government (DILG)	Environmental management is a scored area; LGUs are urged to exceed diversion targets	Proof of diversion for the Seal (DILG, 2024; DILG, 2025)
Baguio Ordinances 15 and 69, s. 2026	20% cut in waste generation; ₱75 monthly household fee	Dashboard of tons diverted and money saved (Cruz, 2026; Matayew, 2026)
Philippines–Singapore carbon agreement	Signed April 2026; list of eligible projects pending	Records built to support carbon credits later (Quek & Kuo, 2026)

III. PROPOSED STARTUP SOLUTION

WasteLedger is a software-as-a-service (SaaS) platform that a city subscribes to. In the proposed model, the team builds and runs the software, and the city staffs the weigh stations (ideally with market or GSO utility workers who already handle this waste), pays suppliers and runs composting. No city has agreed to this yet; until one does, the pilot can run on bridge funding (Section IX).

1. **Register.** Each vendor or farmer group gets a reusable crate with a QR sticker.
2. **Weigh and grade.** Station staff scan the QR code with a phone, weigh the crate on a sealed scale, type the weight, take one photo showing the scale and the QR code, and grade the load: clean, minor contamination or rejected. The app works offline and uploads later.
3. **Pay.** Clean waste earns the city's rate (proposed at ₱0.60 a kilo in Baguio), up to a weekly cap per supplier so no one is rewarded for making more waste. The full rate goes only to market sections and suppliers whose food waste was still thrown out with the garbage during the unpaid baseline weeks; vegetable waste that already went to Irisan earns only a small bonus for quality. The city pays from its own bank account once a balance reaches ₱200; WasteLedger never holds the money.
4. **Compost and check.** The waste goes to Irisan. WasteLedger matches each day's station totals against Irisan's intake records. The city's dashboard shows kilos diverted, active suppliers, payouts and money saved, in a form the city can reuse for its self-compliance reports (NSWMC, 2016).

What a weigh station uses

Item	Purpose
Digital platform scale (100 kg or more), sealed	Weighs each crate; checked weekly against the market's official Timbangan ng Bayan scale (Timbangan ng Bayan Act, 2022)
Android phone with the WasteLedger app	Scans QR codes with its camera and records weight, photo and grade, even offline
QR-tagged crates and waterproof supplier ID cards	Identify the supplier; the app subtracts the empty crate's weight automatically

How the records stay honest

- **Proof for every load:** a photo of the scale display and the crate's QR code in the same frame.
- **Match with the facility:** station totals must agree with what Irisan records receiving.
- **No double counting:** every record gets a unique ID on the phone, so a record uploaded twice after a signal drop still counts once.
- **No silent edits:** records cannot be deleted, and every correction is logged with who made it and when.
- **People decide:** simple AI checks flag unusual loads, such as repeated or padded weights, for staff to review; pay is never withheld automatically.
- **Privacy:** no government ID numbers are collected, and the team will register with the National Privacy Commission (Data Privacy Act of 2012, 2012).

Technology. The core is SaaS: the station app, the payout list and the city dashboard. Behind them is the team's own software: a matching engine that checks station records against the facility's intake records; a rate-rule engine that sets each city's payout so the city always saves; offline-first sync, which saves every record on the phone and uploads it when the phone gets signal, since market signal is often weak; and AI anomaly detection that flags unusual loads for review, starting as simple rules and learning from pilot data. If a partner builds a biodigester, such as the Department of Science and Technology's portable design (Industrial Technology Development Institute [ITDI], 2014), the same records can extend to it: IoT sensors would log the waste that goes in and the gas that comes out. The core pilot does not depend on it.

SDGs. SDG 11.6 (city waste), SDG 12.3 (food loss) and SDG 12.5 (waste reduction); later, SDG 13 (climate action), and SDG 7.2 (clean energy) if a partner digester is built (United Nations, 2015).

IV. OBJECTIVES

General objective. To show that paying suppliers per kilo for clean biodegradable waste, with checked records, moves more market waste out of the landfill haul and costs the city less than hauling it.

When	Goal	Measure of success
During PSC XI	Prototype and validation	Clickable prototype of the app and dashboard; Baguio baseline from GSO data; interviews with 30 vendors and 10 farmers; a letter of interest from Baguio
Pilot, weeks 1–2	Unpaid baseline	How much food waste is still thrown out with the market garbage
Pilot, months 1–3	Participation and quality	20 to 30 vendors and 1 farmer cooperative; at least 70% dropping off weekly; under 5% rejected loads; station totals match Irisan intake
First 6 months	Prove savings	Each station handles at least 1.2 tons a day, so the cost per diverted kilo, including staff time, stays below ₱1.45
First 12 months	Grow composting	Market food waste reaching Irisan grows to at least 10 tons a day (now about 5 to 7)
Year 2	Prove it travels	One Metro Manila city and one Visayas or Mindanao city running the same software with local settings

Targets will be adjusted with partner input.

V. TARGET MARKET/BENEFICIARIES

Cities pay. Vendors, farmers and, later, households are paid and never pay.

Segment	Role	What they gain
Baguio City (GSO, City Environment and Parks Management Office and market administration)	Paying customers, Phase 1	Lower hauling costs and verified diversion data

Segment	Role	What they gain
Baguio public market vendors (about 4,000)	Suppliers, Phase 1	Cash for waste they already produce (INQUIRER.net, 2026)
Benguet farmers who bring unsold produce to Baguio's weigh stations	Suppliers, Phase 1	Income from produce that can no longer be sold
Irisan and other composting or black soldier fly operators	Partners	Cleaner waste and intake records
Large cities with high disposal costs (Quezon City, Manila, Caloocan, Davao City, Cebu City)	Paying customers, Year 2	The same savings at a larger scale
Baguio households	Suppliers, Phase 2	Payouts for segregated food waste
La Trinidad, then other LGUs nationwide	Customers, Phase 2–3	The same platform with local settings
Private generators that pay their own garbage fees (private markets, malls, hotels, restaurants)	Fallback customers	Lower garbage fees; they pay WasteLedger directly

VI. VALUE PROPOSITION

Paying by weight for food waste works abroad: a Jakarta waste bank has paid for food scraps since 2014 (ANTARA, 2019), and South Korea charges households by the weight of their food waste (Lee et al., 2024). WasteLedger brings the idea to Philippine cities and adds what makes public money safe to spend: every paid kilo is weighed, photographed and matched with what the composting facility receives.

- **City:** spends about ₱1.03 to divert a kilo that costs ₱1.45 to haul and dump, and gets records ready for compliance reports and the Seal of Good Local Governance.
- **Vendors and farmers:** a small, visible income for waste they already produce: about ₱12 a day, or roughly ₱300 a month, for a vendor bringing 20 kg a day. It is priced so that selling good produce always pays more, and the vendor interviews will test whether it is enough.
- **Composting facilities:** cleaner waste to compost and a record of what arrived and from whom.

VII. BUSINESS MODEL

WasteLedger sells software to cities (business-to-government). It never buys or resells waste and never holds payout money.

Stream	Who pays	Basis	When
City subscription (main)	LGU	₱300 per extra ton the composting facility confirms receiving (not vegetable waste that already went there), capped at 25% of the city's hauling cost per ton; minimum ₱1,500 per station a month; annual contract, waived during the pilot	Phase 1
Facility tools	Composting partners	Intake and inventory tools	Phase 1
Carbon data	Carbon project developers	Verified diversion data for carbon credits	Later
Digester monitoring	Biodigester operators	Monthly fee per partner digester, if one is built	Later

Rate rule. The payout, the ₱0.30 fee, up to ₱0.13 in transfer fees (Xendit, 2026) and a ₱0.20 minimum saving for the city must together stay at or below the city's hauling cost per kilo. In Baguio: ₱0.60 + ₱0.30 + ₱0.13 + ₱0.20 = ₱1.23, below ₱1.45. If hauling costs fall, the payout is lowered or paid in vouchers. The ₱0.13 is a ceiling: Xendit charges about ₱26 per payout (1%, minimum ₱15, plus ₱11), or about ₱0.08 a kilo at the ₱200 payout threshold. Payouts run through the city's own Xendit account, so WasteLedger never holds the money.

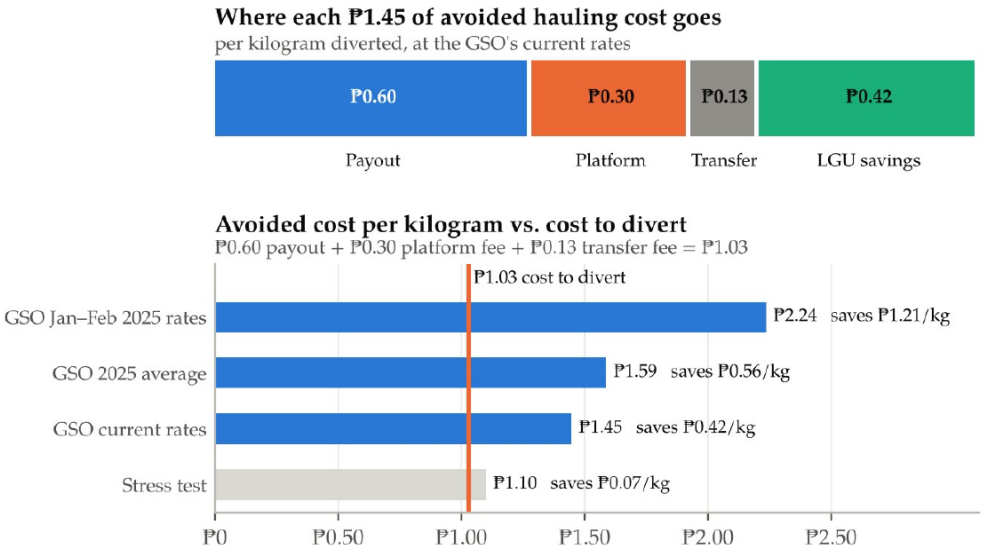


Figure 4. Where each ₱1.45 goes (top), and the cost to divert against the city's hauling cost under four rate scenarios (bottom). In the ₱1.10 stress test, the rate rule cuts the payout to ₱0.47. Source: GSO (2026); team computation.

Baguio-wide savings. These depend on how much of the 12,960 tons of market garbage sent to the landfill is food waste. The GSO says market food waste already goes to Irisan, so the share may be low; the pilot's first two unpaid weeks will measure it.

Share of market garbage that is food waste	Tons diverted a year	Hauling cost avoided	Net saving after payouts and fees, before staff
10%	1,300	₱1.9 million	₱0.5 million
20%	2,600	₱3.7 million	₱1.1 million
35.9% (citywide share)	4,650	₱6.7 million	₱1.9 million

A 500 kg-a-day pilot running six days a week saves the city about ₱5,400 a month before staff costs. A new station worker at ₱505 a day costs about ₱13,100 a month, so a station breaks even at about 1.2 tons a day; at Baguio-wide volume (about 13 tons a day at the 35.9% share) the stations pay for themselves. Source: GSO (2026); team computation.

Carbon credits (later). Checked records are built so they could support carbon credits under the Philippines–Singapore agreement once its list of eligible projects is published (Quek & Kuo, 2026). Carbon income is not counted in this plan.

Procurement. A pilot subscription fits direct acquisition (up to ₱200,000) or small value procurement under the New Government Procurement Act (Government Procurement Policy Board [GPPB], 2025). RA 9003 Section 45 allows LGUs to give incentives for waste reduction, and city payouts need a rate ordinance and a budget line (Ecological Solid Waste Management Act of 2000, 2001; Local Government Code of 1991, 1991).

VIII. MARKET ANALYSIS

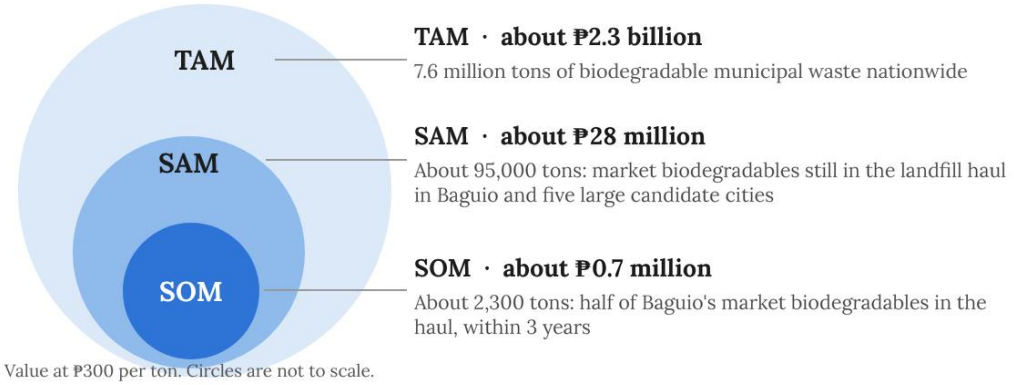


Figure 5. Market size at ₱300 per ton: total addressable market (TAM), serviceable market (SAM) and obtainable market (SOM). The SOM assumes 50% capture in Baguio within three years (team assumption).

Layer	Tons and value	How it is computed
TAM	7.6 million t; about ₱2.3 billion	All municipal waste nationwide, 14.6 million t, × 52% biodegradable (Dela Peña, 2025; Mantaring, 2024)
SAM	about 95,000 t; about ₱28 million	Baguio market food waste in the landfill haul (4,650 t) and five large cities (90,000 t, below). Only additional tons count, so La Trinidad's trading-post waste, already composted at Alno (Sanchez et al., 2019), is excluded
SOM	about 2,300 t; about ₱0.7 million	Half of Baguio's 4,650 t, within three years. Each Year 2 city adds more: Cebu City at 50% is about 4,400 t, or ₱1.3M a year (Section IX)

The five large cities are sized from Baguio's market data: market food waste (2,551 t composted plus an estimated 4,650 t in the haul) are about 3.45% of all the waste Baguio generates. The table applies that share to each city's total waste and assumes none of it is composted yet (Quezon City already composts some), so it is a rough, upper estimate (team assumption):

City	Waste per day (t)	Waste per year (t)	Market food waste (t)	At 50% capture (t)
Quezon City	3,600	1,314,000	45,400	22,700
Manila	1,174	428,510	14,800	7,400
Caloocan	912	332,880	11,500	5,700
Davao City	753	274,845	9,500	4,700
Cebu City	700	255,500	8,800	4,400
Total	**7,139**	**2,605,735**	**90,000**	**45,000**

Figures rounded. At 50% capture, 45,000 tons a year is worth about ₱13.5 million in subscriptions and avoids about ₱65 million in hauling at Baguio's cost (about ₱19 million net of payouts and fees); more in Cebu City.

Competition

Player	What it does	WasteLedger's difference
SM Trash to Cash; Quezon City Trash to Cashback	Cash or rewards for recyclables (SM Supermalls, 2025; Quezon City Government, 2026)	Covers food waste and checks it against the facility
Soilmate Collective (Metro Manila)	Food waste collection paid for by subscribers; over 204 tons diverted from 2021 to early 2024 (Antonio, n.d.)	There the supplier pays; here the city pays suppliers from its hauling savings. A possible partner
TrashTrack	LGU waste monitoring and compliance reports (TrashTrack, 2026)	Works at the supplier level and pays per kilo

Player	What it does	WasteLedger's difference
City composting (Quezon City zero-waste market, Irisan)	Composting without paying vendors (Quezon City Government, n.d.)	Partners and likely customers, not rivals

Edge. WasteLedger owns the paid, checked handoff from supplier to composter. The software can be copied; what lasts is a first reference city, a growing history of matched data and a repeatable rollout.

IX. OPERATIONS PLAN

Phase	When	What happens	Gate to next phase
0. Prototype	During PSC XI	Clickable prototype; GSO baseline; interviews; one observed weighing trial with no payouts	Prototype done; letter of interest
Pilot	Months 1–3	One market section whose food waste still goes out with the garbage, and one farmer cooperative; two unpaid baseline weeks, then payouts	Under 5% contamination; totals match Irisan; savings per kilo shown
1. Baguio-wide	Months 4–12	Whole market, with each station above its break-even of about 1.2 tons a day; subscription starts	A barangay agrees to pilot households
2. More cities	Year 2	One Metro Manila and one Visayas or Mindanao city; Baguio households	Both cities complete a payout cycle
3. National	Year 2–3 onward	More cities through the DILG, procurement and lender programs	Ongoing
Digester monitoring	If a partner builds one	Log the digester's feed and gas output with IoT sensors, using the same records	The partner shares its digester data

What a city needs to join

Requirement	Baguio today
Its hauling and tipping cost per ton	₱1,446.62, from the GSO
A public market that will host a weigh station	Baguio public market
A composting facility that records what it receives	Irisan Composting Facility
A rate ordinance and budget for payouts	To be drafted with the city
A bank account for payouts	Landbank

Where to start and how to grow. The team would next approach cities where disposal costs are rising fastest and a composter already exists, such as Quezon City and Cebu City; no partner is confirmed. To reach many cities, it would use the DILG's push for higher diversion (DILG, 2025), lender-funded city programs such as the World Bank's proposed Clean Metro Manila project (World Bank, 2026), and the procurement routes in Section VII.

If the city cannot fund payouts yet

A city needs a rate ordinance and a budget line before it can pay suppliers, and Baguio's 2027 budget is already being finalized; the mayor submits it by October 16 (Local Government Code of 1991, 1991). So the pilot first asks the city for permission, not money: a memorandum of agreement for station space, Irisan's intake records and data. The pilot's costs can be covered in other ways:

- **Bridge funding:** about ₱26,000 covers 12 weeks of payouts; prize money, a university fund or a company's corporate social responsibility program can cover it.

- **Station staff:** if the city cannot assign staff, a grant pays one or two at the ₱505 Cordillera minimum wage (WageIndicator Foundation, 2025), and team members cover some shifts.
- **Non-cash rewards:** sponsor vouchers, or compost from Irisan for farmers if the GSO allows it.
- **Processor contribution:** a composter or black soldier fly farm that wants clean food waste may share the payout cost.

The pilot's results then support city funding in a later budget. If no city adopts the model, private generators that pay their own garbage fees can use the same software and pay WasteLedger directly.

Key risks

Risk	Response
No city partner confirmed yet	Letters of interest before the regional round; the pilot can run on bridge funding
Hauling costs fall	The rate rule lowers the payout or switches to vouchers, so the city always saves
Little food waste left in the market garbage	Baseline weeks measure it; only newly diverted kilos earn the full rate
Padded or dirty loads	Sealed scales, photos, weekly caps, matching with Irisan and staff review
Other cities take longer to onboard	The pilot records onboarding time and cost first
Carbon income is uncertain	Not counted in this plan

Compliance. Data Privacy Act consent and registration (Data Privacy Act of 2012, 2012); payments only from the city's own bank; procurement under the New Government Procurement Act (2024); startup registration under the Innovative Startup Act (2019).

X. FINANCIAL REQUIREMENT

Item	Low	High	Funding
Station equipment: QR crates, two digital scales, two station phones, labels and signage	₱28,100	₱60,500	Seeks support (prize, grant or sponsor)
Cloud hosting, database and domain	₱3,000	₱7,500	Seeks support (prize, grant or sponsor)
Registration and data privacy compliance, supplier ID cards, gloves, cleaning supplies, scale sealing	₱3,000	₱8,000	Seeks support (prize, grant or sponsor)
Transport and meetings	₱5,000	₱10,000	Team-funded
Station staff: one or two staff for 14 weeks, 6 days a week, at ₱505 a day	₱42,420	₱84,840	City staff if the city agrees; otherwise grant or sponsor, with team members covering some shifts
Supplier payouts and transfer fees: 12 weeks at 500 kg a day, six days a week	₱26,280	₱26,280	Bridge funding until a city budget line exists
Contingency (10%)	₱10,780	₱19,710	Seeks support (prize, grant or sponsor)
Full pilot cost (3 months)	**₱118,580**	**₱216,830**	

Two unpaid baseline weeks plus 12 paid weeks. Phones cost ₱0 if staff phones are used, and staff cost the team ₱0 if the city assigns its own. Staff at ₱505 a day (WageIndicator Foundation, 2025); payouts at 36,000 kg (500 kg a day, six days a week, 12 weeks) × ₱0.60, plus up to ₱0.13 a kilo in transfer fees. Scale prices from KHM Tools (2026); other amounts are team estimates to be replaced with quotes.

Ask. About ₱45,000 to ₱96,000 for equipment, software, registration and contingency, and about ₱26,000 in bridge funding for 12 weeks of supplier payouts; from the city, a letter of interest and a memorandum of

agreement for station space, Irisan's intake records and data; and mentorship on payments, data privacy and LGU procurement.

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